

ROIAlign using Bilinear Interpolation

1.1 Derive the formula for bilinear interpolation

Consider a 2D feature map with corner points:

$$(x, y) \rightarrow f_1, \quad (x+1, y) \rightarrow f_2, \quad (x, y+1) \rightarrow f_3, \quad (x+1, y+1) \rightarrow f_4$$

Let (p, q) be a point such that

$$x \leq p \leq x+1, \quad y \leq q \leq y+1.$$

Define:

$$\alpha = p - x, \quad \beta = q - y, \quad \alpha, \beta \in [0, 1].$$

Then the bilinear interpolation formula is given by:

$$f(p, q) = f_1(1 - \alpha)(1 - \beta) + f_2\alpha(1 - \beta) + f_3(1 - \alpha)\beta + f_4\alpha\beta$$

1.2 Compute partial derivatives

Partial derivative with respect to p

Since $\frac{\partial \alpha}{\partial p} = 1$, and β is independent of p , we have:

$$\frac{\partial f}{\partial p} = (f_2 - f_1)(1 - \beta) + (f_4 - f_3)\beta$$

Partial derivative with respect to f_1

$$\frac{\partial f}{\partial f_1} = (1 - \alpha)(1 - \beta)$$

(Optional) Partial derivative with respect to q

For completeness:

$$\frac{\partial f}{\partial q} = (f_3 - f_1)(1 - \alpha) + (f_4 - f_2)\alpha$$